

Sprint 3 Notes: Text Content and Inline Meaning

Sprint 3 Goal

The goal of Sprint 3 is to learn how to mark up text based on **meaning**, not appearance.

In Sprint 1, you learned how to write individual HTML elements correctly.

In Sprint 2, you learned how to place those elements inside a complete HTML document.

In Sprint 3, you learn how to choose the correct HTML element for different kinds of text.

By the end of this sprint, you should be able to:

- Display reserved HTML characters as normal text.
- Choose between `i` and `em` correctly.
- Choose between `b` and `strong` correctly.
- Build description lists using `dl`, `dt`, and `dd`.
- Mark up long and short quotations.
- Use `cite` correctly.
- Mark abbreviations with `abbr`.
- Mark dates and times with `time`.
- Use `sup` and `sub` for raised and lowered text.
- Mark code using `code` and `pre`.
- Use `u`, `s`, and `ruby` only for their proper meanings.
- Avoid using HTML only for visual styling.

1. Why Sprint 3 Matters

HTML is not only for putting text on a page.

HTML also explains what the text **means**.

For example, HTML can tell the browser:

- This is a paragraph.
- This word is emphasized.
- This text is important.
- This is a warning.
- This is an abbreviation.
- This is a quote.
- This is a date.
- This is code.

- This text is no longer accurate.

A beginner may think:

```
<p>I want this word to look <b>bold</b>.</p>
```

But Sprint 3 asks a better question:

What does this text mean?

If the text is important or urgent, use `strong`.

```
<p><strong>Warning:</strong> This action cannot be undone.</p>
```

If the text only needs attention but is not urgent, use `b`.

```
<p>The product name is <b>SuperSound 3000</b>.</p>
```

Main rule:

Choose HTML elements because of meaning, not because of appearance.

2. Text Meaning vs Text Appearance

Many HTML elements have default browser styles.

For example:

- `em` often looks italic.
- `i` often looks italic.
- `strong` often looks bold.
- `b` often looks bold.
- `u` often looks underlined.
- `s` often looks crossed out.

But these visual styles are not the main reason to use the elements.

The main reason is meaning.

Bad thinking:

```
I want italic text, so I will use em.
```

Better thinking:

```
This word is emphasized, so I will use em.
```

Bad thinking:

```
I want bold text, so I will use strong.
```

Better thinking:

```
This warning is important, so I will use strong.
```

Use CSS later when you only want to change appearance.

Use HTML when you want to describe meaning.

3. HTML Entities

What Are HTML Entities?

Some characters have special meaning in HTML.

For example:

```
<p>Hello</p>
```

The browser reads `<p>` as an HTML paragraph tag.

But sometimes you want to show `<p>` as plain text on the page.

To do that, you use an **HTML entity**.

An HTML entity is a special code that displays a reserved character.

Example:

```
<p>This is a paragraph tag: &lt;p&gt;</p>
```

The browser displays:

```
This is a paragraph tag: <p>
```

Common HTML Entities

Entity	Displays As	Meaning
<code>&lt;</code>	<code><</code>	Less-than sign
<code>&gt;</code>	<code>></code>	Greater-than sign
<code>&amp;</code>	<code>&</code>	Ampersand
<code>&copy;</code>	<code>©</code>	Copyright symbol

Examples

```
<p>Use &lt;h1&gt; for the main page heading.</p>  
<p>&copy; 2026 Example Company</p>  
<p>Use &amp; when writing an ampersand as text.</p>
```

Why Entities Matter

If you write this:

```
<p>This is an <img> tag.</p>
```

The browser may treat `<img>` as an actual image element.

That is not what you want.

Correct version:

```
<p>This is an &lt;img&gt; tag.</p>
```

Rule

Use entities when you want the browser to display characters that HTML might otherwise interpret as code.

4. `i` vs `em`

Both `i` and `em` often look italic in the browser.

But they do not mean the same thing.

The `i` Element

Use `i` when text is in a different voice, mood, or mode.

Good uses for `i` include:

- Foreign phrases.
- Technical terms.
- Idioms.
- Thoughts.
- Alternative voice.
- Names of ships or similar special terms.

Example:

```
<p>There is a certain <i lang="fr">je ne sais quoi</i> in the design.</p>
```

Here, the phrase `je ne sais quoi` is French.

The `i` element shows that the phrase is in a different voice or language.

The `lang="fr"` attribute tells the browser and assistive technology that this phrase is French.

Another example:

```
<p>The term <i>semantic HTML</i> means HTML that describes meaning.</p>
```

Here, `semantic HTML` is being treated as a technical term.

The `em` Element

Use `em` when the word is emphasized.

Emphasis means the stress changes the meaning of the sentence.

Example:

```
<p>I said you should submit <em>today</em>, not tomorrow.</p>
```

The word `today` is stressed.

The sentence means something different because of that stress.

Another example:

```
<p>You must write the code <em>yourself</em>.</p>
```

The emphasis is on `yourself`.

`i` vs `em` Simple Rule

```
i = different voice or mode  
em = stressed meaning
```

Use `i` when the text is different from the surrounding text.

Use `em` when the text needs stress.

Common Mistake

Incorrect:

```
<p>This word is <em>italic</em>.</p>
```

Why this is weak:

The reason is appearance, not meaning.

Better:

```
<p>I said finish it <em>today</em>.</p>
```

Now `em` is used because the word has emphasis.

5. `b` vs `strong`

Both `b` and `strong` often look bold in the browser.

But they do not mean the same thing.

The `b` Element

Use `b` to bring attention to text without saying it is urgent or important.

Good uses for `b` include:

- Product names.
- Keywords.
- Names that should stand out.
- Text that needs attention but not importance.

Example:

```
<p>We tested the <b>SuperSound 3000</b>.</p>
```

Here, the product name is being highlighted.

It is not a warning.

It is not urgent.

It only needs attention.

The `strong` Element

Use `strong` for important, serious, or urgent text.

Example:

```
<p><strong>Warning:</strong> Do not share your password.</p>
```

Here, the text is important.

Another example:

```
<p><strong>Important:</strong> The deadline is today.</p>
```

Use `strong` when the meaning is stronger than normal text.

`b` vs `strong` Simple Rule

```
b      = attention
strong = importance or urgency
```

Use `b` for text that should stand out.

Use `strong` for text that the user must take seriously.

Common Mistake

Weak:

```
<p><strong>SuperSound 3000</strong> is our new product.</p>
```

Why this is weak:

The product name is not necessarily important or urgent.

Better:

```
<p><b>SuperSound 3000</b> is our new product.</p>
```

Correct use of `strong`:

```
<p><strong>Warning:</strong> This product may cause allergic reactions.</p>
```

6. Description Lists: `dl`, `dt`, and `dd`

What Is a Description List?

A description list is a list of terms and details.

It is useful when your content has pairs such as:

- Word and definition.
- Question and answer.
- Product feature and value.
- Name and description.
- Setting and explanation.
- Contact detail and value.

The Three Elements

A description list uses three main elements:

```
dl = description list
dt = description term
dd = description detail
```

Basic structure:

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>

  <dt>CSS</dt>
  <dd>Cascading Style Sheets</dd>
</dl>
```

Meaning:

- `dl` wraps the whole description list.
- `dt` gives the term or name.
- `dd` gives the explanation or detail.

Good Uses

Description lists are good for glossaries.

```
<dl>
  <dt>HTML</dt>
  <dd>The language used to structure web pages.</dd>
```

```
<dt>CSS</dt>
<dd>The language used to style web pages.</dd>

<dt>JavaScript</dt>
<dd>The language used to add behavior to web pages.</dd>
</dl>
```

Description lists are also good for product specs.

```
<dl>
  <dt>Screen size</dt>
  <dd>15 inches</dd>

  <dt>Memory</dt>
  <dd>16 GB</dd>

  <dt>Storage</dt>
  <dd>512 GB SSD</dd>
</dl>
```

Weak Version

```
<p>HTML: HyperText Markup Language</p>
<p>CSS: Cascading Style Sheets</p>
```

This works visually, but it does not describe the relationship clearly.

Better Version

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>

  <dt>CSS</dt>
  <dd>Cascading Style Sheets</dd>
</dl>
```

This gives the content better structure and meaning.

7. Quotations: `blockquote`, `q`, `cite`, and `cite` Attribute

HTML has specific elements for quotations and source information.

The `blockquote` Element

Use `blockquote` for a longer quote.

Example:

```
<blockquote cite="https://example.com/html-guide">
  <p>Good HTML describes the structure and meaning of content.</p>
</blockquote>
```

The `blockquote` element means this is a separate quoted section.

The `cite` attribute can hold the source URL.

Important:

The `cite` attribute is not usually shown visibly on the page.

It gives machine-readable source information.

The `q` Element

Use `q` for a short inline quote.

Example:

```
<p>The teacher said <q>semantic HTML matters</q> during the lesson.</p>
```

The quote is inside a normal sentence.

That is why `q` is better than `blockquote` here.

The `cite` Element

Use the `cite` element for the title of a work.

A work can be:

- A book.
- An article.
- A guide.
- A film.
- A report.
- A website resource.

Example:

```
<p>This idea appears in <cite>HTML Beginner Guide</cite>.</p>
```

The visible text is the title of the source.

`<cite>` Element vs `<cite>` Attribute

These are different.

```
<blockquote cite="https://example.com/article">
  <p>This is a long quote.</p>
</blockquote>
```

Here, `<cite>` is an attribute.

It gives the source URL.

```
<p>Source: <cite>HTML Beginner Guide</cite></p>
```

Here, `<cite>` is an element.

It marks the title of a work.

Simple Rule

```
blockquote = long quote
q          = short inline quote
cite       = title of a source or work
cite=""    = source URL attribute
```

8. Abbreviations: abbr

What Is abbr?

Use `abbr` for abbreviations or acronyms.

Example:

```
<p><abbr title="HyperText Markup Language">HTML</abbr> structures web pages.</p>
```

The visible text is:

HTML

The full meaning is in the `title` attribute:

HyperText Markup Language

Why `abbr` Is Useful

The abbreviation may be clear to you, but not to every reader.

The `title` attribute gives the full meaning.

It helps explain the abbreviation.

Correct Pattern

```
<abbr title="HyperText Markup Language">HTML</abbr>
```

The short form goes between the opening and closing tags.

The full form goes in the `title` attribute.

Common Mistake

Weak:

```
<p>HTML means <abbr>HyperText Markup Language</abbr>.</p>
```

Why this is weak:

The abbreviation is missing.

Better:

```
<p><abbr title="HyperText Markup Language">HTML</abbr> means HyperText Markup Language.</p>
```

9. Dates and Times: `time`

What Is `time`?

Use the `time` element when your text contains a date, time, or duration.

Example:

```
<p>The event is on <time datetime="2026-05-07">May 7, 2026</time>.</p>
```

The visible text is:

```
May 7, 2026
```

The machine-readable value is:

```
2026-05-07
```

The machine-readable value goes in the `datetime` attribute.

Why `datetime` **Matters**

Humans can understand many date formats.

For example:

May 7, 2026
7 May 2026
Thursday, May 7

But computers need a clearer format.

The `datetime` attribute gives a precise machine-readable date or time.

Date Example

```
<p>Published on <time datetime="2026-05-07">May 7, 2026</time>.</p>
```

Date and Time Example

```
<p>The meeting starts at <time datetime="2026-05-07T15:30">May 7 at 3:30 PM</time>.</p>
```

The `T` separates the date from the time.

```
2026-05-07T15:30
```

Means:

```
May 7, 2026 at 15:30
```

Simple Rule

Use `time` when you mention a meaningful date or time.

Use `datetime` to give the browser a clean machine-readable value.

10. Superscript and Subscript: `sup` and `sub`

The `sup` Element

Use `sup` for superscript text.

Superscript text appears raised above the normal text line.

Example:

```
<p>2<sup>2</sup> = 4</p>
```

This displays like:

```
22 = 4
```

Good uses for `sup` include:

- Exponents.
- Footnote markers.
- Ordinals.
- Superior lettering.

Example:

```
<p>The 1<sup>st</sup> place winner received a medal.</p>
```

The `sub` Element

Use `sub` for subscript text.

Subscript text appears lowered below the normal text line.

Example:

```
<p>C0<sub>2</sub> is carbon dioxide.</p>
```

This displays like:

CO₂ is carbon dioxide.

Good uses for `sub` include:

- Chemical formulas.
- Mathematical subscripts.
- Footnote-style references in some contexts.

Another example:

```
<p>H<sub>2</sub>O is water.</p>
```

Simple Rule

```
sup = raised text  
sub = lowered text
```

Use `sup` and `sub` when the raised or lowered position has meaning.

Do not use them only for decoration.

11. Code Text: `code` and `pre`

HTML has elements for code examples.

The `code` Element

Use `code` for short inline code.

Example:

```
<p>Use the <code>p</code> element for paragraphs.</p>
```

Here, `p` is code text inside a normal sentence.

Another example:

```
<p>The CSS property <code>color</code> changes text color.</p>
```

The `pre` Element

Use `pre` when spacing and line breaks must be preserved.

The browser normally collapses extra spaces and line breaks.

The `pre` element keeps spacing exactly as written.

Example:

```
<pre>This text
keeps its line breaks
and spacing.</pre>
```

Using `pre` and `code` Together

For multi-line code blocks, use `pre` with `code` inside it.

Example:

```
<pre><code>body {
  color: red;
}</code></pre>
```

Meaning:

```
code = this is code
pre  = preserve spacing and line breaks
```

HTML Code Example

If you want to show HTML code as text, you need entities.

```
<pre><code>&lt;p&gt;This is a paragraph.&lt;/p&gt;
&lt;strong&gt;Important text&lt;/strong&gt;</code></pre>
```

The browser displays the tags as text instead of treating them as real HTML elements.

Common Mistake

Weak:

```
<pre>
  <p>This should be code</p>
</pre>
```

Problem:

The browser may treat `<p>` as a real paragraph element.

Better:

```
<pre><code>&lt;p&gt;This should be code&lt;/p&gt;</code></pre>
```

12. The `u` Element

What Is `u` ?

The `u` element represents non-textual annotation.

In simple words, it marks text that has a special annotation, such as a spelling issue.

Example:

```
<p><u>speling</u> error</p>
```

The word `speling` is marked as having an issue.

What `u` Is Not For

Do not use `u` only because you want underlined text.

Weak:

```
<p><u>This text should look underlined.</u></p>
```

Why weak:

That is appearance, not meaning.

Use CSS later if you only want underline styling.

Simple Rule

```
u = non-textual annotation
```

Use `u` only when the annotation has meaning.

13. The `s` Element

What Is `s`?

The `s` element marks text that is no longer accurate or no longer relevant.

Example:

```
<p><s>The sale ends tomorrow.</s></p>
<p>The sale has ended.</p>
```

This means the old statement is no longer true.

What `s` Is Not For

Do not use `s` for formal document edits.

For formal inserted and deleted content, HTML has other elements such as `del` and `ins`, which come later.

Simple Rule

s = no longer accurate or no longer relevant

Use `s` when the content should remain visible but should be understood as outdated or incorrect.

14. Ruby Annotations: `ruby`, `rt`, and `rp`

Ruby annotations are used for pronunciation or extra reading help.

They are common in East Asian typography.

The Three Elements

`ruby` = wraps the base text and pronunciation annotation
`rt` = ruby text, usually pronunciation
`rp` = fallback parentheses for browsers that do not support ruby

Example:

```
<ruby>
  明日 <rp>(</rp><rt>Ashita</rt><rp>)</rp>
</ruby>
```

Meaning:

- 明日 is the base text.
 - Ashita is the pronunciation.
 - `rp` provides fallback parentheses.
-

Beginner Note

You may not use `ruby` often at this stage.

For Sprint 3, you should mainly recognize what it is for.

Use it when text needs pronunciation guidance.

15. Full Sprint 3 Example Page

This example combines Sprint 1, Sprint 2, and Sprint 3 ideas.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>HTML Text Meaning</title>
  <meta name="description" content="A beginner HTML page demonstrating text
meaning and inline semantic elements.">
</head>
<body>
  <h1>HTML Text Meaning</h1>

  <p>
    HTML is not only about displaying text. It also tells the browser what the
    text means.
  </p>

  <p>
    For example, write &lt;p&gt; when you want to show a paragraph tag as text.
  </p>

  <p>
    The phrase <i lang="fr">je ne sais quoi</i> is written in a different
    language.
  </p>

  <p>
    You should submit the assignment <em>today</em>, not tomorrow.
  </p>

  <p>
    The product name is <b>CodePad Mini</b>.
  </p>

  <p>
    <strong>Warning:</strong> Do not share your password with anyone.
  </p>

  <h2>Glossary</h2>

  <dl>
    <dt><abbr title="HyperText Markup Language">HTML</abbr></dt>
```

```

<dd>The language used to structure content on a web page.</dd>

<dt>Entity</dt>
<dd>A special code used to display reserved characters like &lt; and &gt;.</dd>
dd>

<dt>Semantic HTML</dt>
<dd>HTML that describes the meaning of content.</dd>
</dl>

<h2>Quote Example</h2>

<blockquote cite="https://example.com/html-guide">
  <p>Good HTML describes the structure and meaning of content.</p>
</blockquote>

<p>
  Many developers say <q>semantic HTML improves clarity</q>.
</p>

<p>
  Source: <cite>HTML Beginner Guide</cite>
</p>

<h2>Date and Technical Text</h2>

<p>
  This lesson was written on <time datetime="2026-05-07">May 7, 2026</time>.
</p>

<p>
  In chemistry, CO<sub>2</sub> means carbon dioxide.
</p>

<p>
  In math, 2<sup>3</sup> equals 8.
</p>

<p>
  Use <code>&lt;p&gt;</code> for a paragraph.
</p>

<pre><code>&lt;p&gt;This is a paragraph.&lt;/p&gt;
&lt;strong&gt;This is important.&lt;/strong&gt;</code></pre>

<p>
  <s>This lesson is incomplete.</s> This lesson now includes Sprint 3
  examples.

```

```
</p>
</body>
</html>
```

16. Explanation of the Full Example

Boilerplate

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>HTML Text Meaning</title>
</head>
<body>
  ...
</body>
</html>
```

This comes from Sprint 2.

It creates a complete HTML document.

Main Heading

```
<h1>HTML Text Meaning</h1>
```

This gives the page its main visible heading.

Entity Example

```
<p>
  For example, write &lt;p&gt; when you want to show a paragraph tag as text.
</p>
```

`<` displays `<`.

`>` displays `>`.

This allows the page to show `<p>` as text.

`i` Example

```
<i lang="fr">je ne sais quoi</i>
```

This phrase is French, so it is marked as a different language and voice.

`em` Example

```
<em>today</em>
```

The word `today` is emphasized.

The stress changes the meaning of the sentence.

`b` Example

```
<b>CodePad Mini</b>
```

The product name gets attention.

It is not a warning or urgent message.

`strong` Example

```
<strong>Warning:</strong>
```

This marks important warning text.

Description List Example

```
<dl>
  <dt>Entity</dt>
  <dd>A special code used to display reserved characters like &lt; and &gt;.</
```

```
dd>  
</dl>
```

This shows a term and its explanation.

Quote Example

```
<blockquote cite="https://example.com/html-guide">  
  <p>Good HTML describes the structure and meaning of content.</p>  
</blockquote>
```

This is a long quotation.

The `<cite>` attribute gives the source URL.

Inline Quote Example

```
<q>semantic HTML improves clarity</q>
```

This is a short quote inside a sentence.

Cite Element Example

```
<cite>HTML Beginner Guide</cite>
```

This marks the title of a source.

Time Example

```
<time datetime="2026-05-07">May 7, 2026</time>
```

The visible date is human-readable.

The `<datetime>` value is machine-readable.

Subscript Example

```
CO<sub>2</sub>
```

The `2` appears lower because chemical formulas use subscript.

Superscript Example

```
2<sup>3</sup>
```

The `3` appears higher because exponents use superscript.

Inline Code Example

```
<code>&lt;p&gt;</code>
```

This marks `<p>` as code inside a sentence.

Code Block Example

```
<pre><code>&lt;p&gt;This is a paragraph.&lt;/p&gt;
&lt;strong&gt;This is important.&lt;/strong&gt;</code></pre>
```

`code` means the text is code.

`pre` preserves the line breaks and spacing.

`s` Example

```
<s>This lesson is incomplete.</s>
```

This marks old text as no longer accurate.

17. Common Sprint 3 Mistakes and Fixes

Mistake 1: Showing Tags Without Entities

Incorrect:

```
<p>This is an <img> tag.</p>
```

Problem:

The browser sees `<img>` as an actual HTML element.

Correct:

```
<p>This is an &lt;img&gt; tag.</p>
```

Why correct:

The browser now displays `<img>` as text.

Mistake 2: Using `strong` for Product Names

Weak:

```
<p><strong>SuperSound 3000</strong> is the product name.</p>
```

Problem:

A product name usually needs attention, not urgent importance.

Better:

```
<p><b>SuperSound 3000</b> is the product name.</p>
```

Mistake 3: Using `abbr` Incorrectly

Weak:

```
<p>HTML means <abbr>HyperText Markup Language</abbr>.</p>
```

Problem:

The abbreviation itself is missing from the `abbr` element.

Better:

```
<p><abbr title="HyperText Markup Language">HTML</abbr> means HyperText Markup Language.</p>
```

Mistake 4: Using `pre` Without `code` for Code

Weak:

```
<pre>
  <p>This should be code</p>
</pre>
```

Problems:

- The text is not marked as code.
- `<p>` may be treated as real HTML.

Better:

```
<pre><code>&lt;p&gt;This should be code&lt;/p&gt;</code></pre>
```

Mistake 5: Using `em` Only for Italic Styling

Weak:

```
<p>This word should look <em>italic</em>.</p>
```

Problem:

The reason is visual appearance.

Better:

```
<p>I said finish it <em>today</em>.</p>
```

Now the word has real emphasis.

Mistake 6: Using `<u>` Only for Underline Styling

Weak:

```
<p><u>This is underlined text.</u></p>
```

Problem:

The underline is only visual.

Better:

```
<p><u>speling</u> error</p>
```

Now the `<u>` element marks a meaningful annotation.

Mistake 7: Using `<s>` Only for Decoration

Weak:

```
<p><s>This text looks cool.</s></p>
```

Problem:

The strikethrough has no meaning.

Better:

```
<p><s>The event starts at 10 AM.</s></p>  
<p>The event now starts at 11 AM.</p>
```

Now the old text is marked as no longer accurate.

18. Sprint 3 Debugging Practice

Broken Code

```
<p>This is an <img> tag.</p>
<p><strong>This product name</strong> is SuperSound 3000.</p>
<p>HTML means <abbr>HyperText Markup Language</abbr></p>
<pre>
  <p>This should be code</p>
</pre>
```

Fixed Code

```
<p>This is an &lt;img&gt; tag.</p>
<p><b>SuperSound 3000</b> is the product name.</p>
<p><abbr title="HyperText Markup Language">HTML</abbr> means HyperText Markup
Language.</p>
<pre><code>&lt;p&gt;This should be code&lt;/p&gt;</code></pre>
```

Explanation of Fixes

Fix 1

Broken:

```
<p>This is an <img> tag.</p>
```

Fixed:

```
<p>This is an &lt;img&gt; tag.</p>
```

Why:

Use entities to display `<img>` as text.

Fix 2

Broken:

```
<p><strong>This product name</strong> is SuperSound 3000.</p>
```

Fixed:

```
<p><b>SuperSound 3000</b> is the product name.</p>
```

Why:

The product name needs attention, not urgent importance.

Fix 3

Broken:

```
<p>HTML means <abbr>HyperText Markup Language</abbr></p>
```

Fixed:

```
<p><abbr title="HyperText Markup Language">HTML</abbr> means HyperText Markup Language.</p>
```

Why:

`abbr` should wrap the abbreviation.

The full phrase belongs in the `title` attribute.

Fix 4

Broken:

```
<pre>
  <p>This should be code</p>
</pre>
```


Fixed:

```
<pre><code>&lt;p&gt;This should be code&lt;/p&gt;</code></pre>
```

Why:

Use `code` for code meaning.

Use `pre` to preserve formatting.

Use entities to display the HTML tags as text.

19. Sprint 3 Practice Task

Create a file named:

```
sprint-3-text-meaning.html
```

Build a page called:

```
HTML Text Meaning
```

Requirements

Your page should include:

- A full Sprint 2 boilerplate.
- One `h1`.
- At least two `p` elements.
- One entity example using `<` and `>`.
- One `i` element.
- One `em` element.
- One `b` element.
- One `strong` warning.
- One `dl` with at least three `dt` and `dd` pairs.
- One `abbr` with a `title` attribute.
- One `blockquote`.
- One `q`.
- One `cite` element.
- One `time` element with `datetime`.

- One `sup` example.
 - One `sub` example.
 - One inline `code` example.
 - One `pre` and `code` block.
 - Optional: one `u`, one `s`, and one `ruby` example.
-

20. Starter Practice Solution

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>HTML Text Meaning</title>
  <meta name="description" content="A Sprint 3 practice page about meaningful
HTML text elements.">
</head>
<body>
  <h1>HTML Text Meaning</h1>

  <p>
    HTML tells the browser what content means, not only how it should appear.
  </p>

  <p>
    To show a paragraph tag as text, write &lt;p&gt; instead of typing the real
tag directly.
  </p>

  <p>
    The phrase <i lang="fr">bon voyage</i> is French.
  </p>

  <p>
    Please finish this exercise <em>before</em> moving to Sprint 4.
  </p>

  <p>
    The example product is called <b>TextPro 100</b>.
  </p>

  <p>
    <strong>Warning:</strong> Do not use HTML elements only for visual styling.
  </p>
```

```
<h2>Glossary</h2>
```

```
<dl>
  <dt><abbr title="HyperText Markup Language">HTML</abbr></dt>
  <dd>The language used to structure content on the web.</dd>

  <dt>Entity</dt>
  <dd>A special code that displays reserved characters.</dd>

  <dt>Semantic element</dt>
  <dd>An element that describes the meaning of content.</dd>
</dl>
```

```
<h2>Quotes</h2>
```

```
<blockquote cite="https://example.com/semantic-html">
  <p>Semantic HTML makes content easier to understand.</p>
</blockquote>
```

```
<p>
  A common rule is <q>choose elements for meaning</q>.
</p>
```

```
<p>
  Source: <cite>Beginner HTML Notes</cite>
</p>
```

```
<h2>Dates, Science, Math, and Code</h2>
```

```
<p>
  This page was created on <time datetime="2026-05-07">May 7, 2026</time>.
</p>
```

```
<p>
  Water is written as H<sub>2</sub>O.
</p>
```

```
<p>
  The expression 5<sup>2</sup> means 5 squared.
</p>
```

```
<p>
  Use <code>&lt;strong&gt;</code> for important text.
</p>
```

```
<pre><code>&lt;p&gt;This is a paragraph.&lt;/p&gt;
&lt;em&gt;This word is emphasized.&lt;/em&gt;</code></pre>
```

```
<p>
  <u>error</u> is marked as a spelling problem.
</p>

<p>
  <s>This note is unfinished.</s> This note is now complete.
</p>

<ruby>
  明日 <rp></rp><rt>Ashita</rt><rp></rp>
</ruby>
</body>
</html>
```

21. Sprint 3 Revision Questions

Use these questions to test yourself.

Question 1

What is the main goal of Sprint 3?

Answer:

Sprint 3 teaches you to mark up text based on meaning, not appearance.

Question 2

Why do we use `<` and `>`?

Answer:

We use them to display `<` and `>` as text instead of letting the browser treat them as HTML syntax.

Question 3

When should you use `i`?

Answer:

Use `i` for text in a different voice or mode, such as a foreign phrase, technical term, idiom, or thought.

Question 4

When should you use `em`?

Answer:

Use `em` when stressing the word changes the meaning of the sentence.

Question 5

When should you use `b`?

Answer:

Use `b` when text needs attention but is not urgent or important.

Question 6

When should you use `strong`?

Answer:

Use `strong` for important, serious, or urgent text.

Question 7

What three elements make a description list?

Answer:

`dl`, `dt`, and `dd`.

Question 8

What does `dl` do?

Answer:

`dl` wraps the whole description list.

Question 9

What does `dt` do?

Answer:

`dt` marks the term or name.

Question 10

What does `dd` do?

Answer:

`dd` marks the description or detail.

Question 11

When should you use `blockquote`?

Answer:

Use `blockquote` for longer quotes that stand apart from the sentence.

Question 12

When should you use `q`?

Answer:

Use `q` for short inline quotes inside a sentence.

Question 13

What does the `cite` attribute do?

Answer:

The `cite` attribute can point to the source URL of a quote.

Question 14

What does the `cite` element do?

Answer:

The `cite` element marks the title of a work, such as a book, article, guide, or resource.

Question 15

How should you write an abbreviation for HTML?

Answer:

```
<abbr title="HyperText Markup Language">HTML</abbr>
```

Question 16

Why does `time` use `datetime`?

Answer:

The visible text is for humans, but the `datetime` value gives a clean machine-readable date or time.

Question 17

When should you use `sup`?

Answer:

Use `sup` for raised text, such as exponents, footnote markers, and ordinals.

Question 18

When should you use `sub`?

Answer:

Use `sub` for lowered text, such as chemical formulas or mathematical subscripts.

Question 19

When should you use `code`?

Answer:

Use `code` for code text, especially short inline code inside a sentence.

Question 20

When should you combine `pre` and `code`?

Answer:

Use `pre` with `code` for multi-line code blocks where spacing and line breaks matter.

Question 21

When should you use `u`?

Answer:

Use `u` for non-textual annotation, such as marking a spelling issue.

Question 22

When should you use `s`?

Answer:

Use `s` for text that is no longer accurate or no longer relevant.

Question 23

What are `ruby`, `rt`, and `rp` used for?

Answer:

They are used for pronunciation annotations, often in East Asian typography.

22. Sprint 3 Key Rules

Memorize these rules:

- Choose HTML elements for meaning, not appearance.
- Use entities to display reserved HTML characters.
- Use `i` for a different voice or mode.
- Use `em` for emphasis that changes meaning.
- Use `b` for attention.
- Use `strong` for importance or urgency.
- Use `dl`, `dt`, and `dd` for description lists.
- Use `blockquote` for long quotes.
- Use `q` for short inline quotes.
- Use the `cite` attribute for a quote source URL.
- Use the `cite` element for the title of a work.
- Use `abbr` for abbreviations.
- Put the full abbreviation meaning in the `title` attribute.
- Use `time` for dates and times.
- Use `datetime` for machine-readable dates and times.
- Use `sup` for raised text with meaning.
- Use `sub` for lowered text with meaning.
- Use `code` for code.
- Use `pre` when spacing and line breaks must be preserved.
- Use `pre` and `code` together for multi-line code blocks.
- Use `u` for annotations, not decoration.
- Use `s` for text that is no longer accurate.
- Use `ruby`, `rt`, and `rp` for pronunciation annotations.
- Use CSS later for appearance-only changes.

23. Sprint 3 Mini Glossary

Entity

A special code used to display reserved characters as text.

Example:

```
&lt;
```

Displays:

```
<
```

i

An element for text in a different voice or mode.

Example:

```
<i lang="fr">bonjour</i>
```

em

An element for emphasized text.

Example:

```
<em>today</em>
```

b

An element for text that needs attention.

Example:

```
<b>SuperSound 3000</b>
```

strong

An element for important or urgent text.

Example:

```
<strong>Warning:</strong>
```

dl

A description list.

`dt`

A description term.

`dd`

A description detail.

`blockquote`

An element for longer quotes.

`q`

An element for short inline quotes.

`cite` **Element**

An element for the title of a source or work.

`cite` **Attribute**

An attribute that can point to the source URL of a quote.

`abbr`

An element for abbreviations.

`time`

An element for dates, times, or durations.

`datetime`

An attribute that gives a machine-readable date or time.

sup

An element for raised text.

sub

An element for lowered text.

code

An element for code text.

pre

An element that preserves spaces and line breaks.

u

An element for non-textual annotation.

s

An element for text that is no longer accurate or relevant.

ruby

An element for pronunciation annotations.

rt

Ruby text, usually pronunciation.

rp

Fallback parentheses for ruby annotations.

24. Sprint 3 Final Self-Check

You are ready to move on from Sprint 3 when you can do these without looking at notes:

- Explain why Sprint 3 focuses on meaning, not appearance.
- Display `<p>` as text using entities.
- Explain the difference between `i` and `em`.
- Explain the difference between `b` and `strong`.
- Build a description list with `dl`, `dt`, and `dd`.
- Use `blockquote` for a long quote.
- Use `q` for an inline quote.
- Use the `cite` attribute for a quote source URL.
- Use the `cite` element for the title of a work.
- Use `abbr title="..."` correctly.
- Use `time datetime="..."` correctly.
- Write a superscript example using `sup`.
- Write a subscript example using `sub`.
- Use `code` for inline code.
- Use `pre` and `code` together for a code block.
- Explain why `u` is not just for underline styling.
- Explain why `s` is not just for strikethrough styling.
- Recognize the purpose of `ruby`, `rt`, and `rp`.
- Fix common Sprint 3 mistakes.

25. Sprint 3 Completion Standard

Sprint 3 is complete when you can write text-focused HTML that communicates meaning clearly.

You should be able to look at text and decide which element fits its purpose.

The final mental model is:

Do not ask: "How do I make this look?"
Ask: "What does this text mean?"

That is the main point of Sprint 3.